

All Courses Course Agentic AI

Course Intermediate 7 hours 45 mins

Agentic AI

Instructor: Andrew Ng

DeepLearning.AI

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What you'll learn

- ✓ **Build agentic design patterns:** reflection, tool use, planning, and multi-agent workflows
- ✓ **Integrate AI with external tools:** databases, APIs, web search, and code execution
- ✓ **Evaluate and optimize AI systems:** performance metrics, error analysis, and production deployment

Agentic AI

- 📈 Intermediate
- 🕒 7 hours 45 mins
- ▶ 31 Video Lessons
- 📄 7 Code Examples
- 📝 8 Graded Assignments PRO
- ✓ Earn a certificate with PRO
- 👤 Instructor: Andrew Ng

Agentic AI represents a new way of building software that leverages LLMs to complete some or all of the steps in complex tasks. Instead of generating single responses to prompts, agentic workflows enable AI to plan multi-step processes, execute them iteratively, and improve outputs through reflection and tool use. This course teaches you to build these sophisticated AI systems from the ground up.

You'll master four design patterns that power agentic AI systems:

- **Reflection:** AI critiques its own work and iterates to improve quality—like code review, but automated.
- **Tool Use:** Connect AI to databases, APIs, and external services so it can actually perform actions, not just generate text.
- **Planning:** Break complex tasks into executable steps that AI can follow and adapt when things don't go as expected.
- **Multi-Agent:** Coordinate multiple specialized AI systems to handle different parts of a complex workflow.

The course emphasizes practical implementation using Python, building each pattern from first principles before exploring frameworks, giving you the flexibility to customize AI workflows for your specific needs. You'll learn to deconstruct business processes into agentic workflows, identifying where human-like iteration and tool interaction can automate complex tasks. Critical evaluation skills are woven throughout—you'll build robust testing frameworks, conduct systematic error analysis, and optimize systems for production deployment.

You'll earn a certificate upon completing the course, recognizing your skills in building agentic workflows.

[Membership PRO Plan](#) 

Start Learning

Instructor



Andrew Ng

Founder, DeepLearning.AI; Co-founder, Coursera



Who this course is for

- ✓ Software developers who want to apply AI techniques to build autonomous systems that handle multi-step workflows.
- ✓ Professionals with intermediate-level Python programming skills to be able to follow along with the implementations.
- ✓ Professionals with a basic understanding of large language models and APIs who want to deepen their practical skills.

Course Outline

Agentic AI

Module 1: Introduction to Agentic Workflows



Module 2: Reflection Design Pattern



Module 3: Tool use

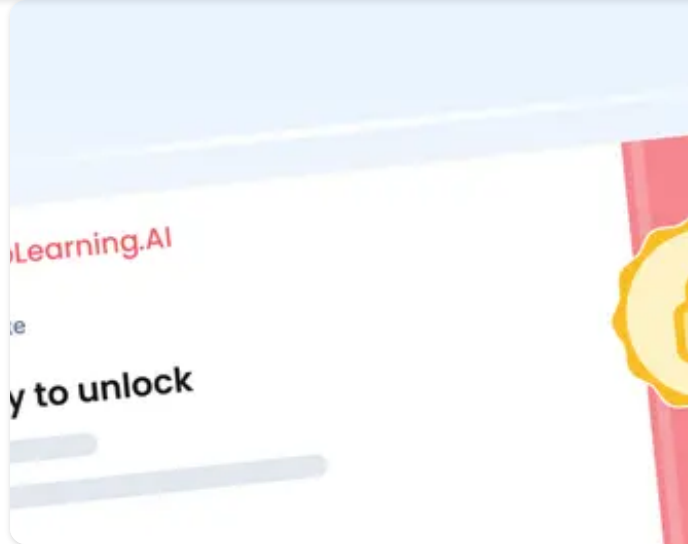


Module 4: Practical Tips for Building Agentic AI



Module 5: Patterns for Highly Autonomous Agents





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Frequently Asked Questions

What additional features do I get with the DeepLearning.AI Pro membership for this course?



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